Conceptual Mapping: Technology (Phases 1–4)

Te Tāhuhu o te Mātauranga
Ministry of Education

Strands	Sub-elements	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6
		Students develop fit-for-purpose technological outcomes through design, making, and innovation					
Design, Make, and Innovate	**Design**	**Technology as purposeful design** Technologists create outcomes to meet needs and solve problems. They develop and communicate ideas, consider the impacts of outcomes, and evaluate how well outcomes achieve their intended purpose.			**Technology as design decision-making** Technologists make design decisions that respond to needs and opportunities. They consider impacts on people and environments, use feedback and evidence to refine ideas, and evaluate fitness for purpose.		
	Materials and ingredients	**Materials and ingredients as resources for creating outcomes** Materials and ingredients have properties that influence how they can be transformed and used to create technological outcomes.			**Materials and ingredients as design decisions** Selecting and processing materials and ingredients involves balancing performance, purpose, impacts, and sustainability.		
	Computational thinking			**Solve problems through ordered thinking** Problems can be solved by breaking them into steps, identifying patterns, and refining solutions.	**Managing complexity through computational thinking** Computational thinking helps manage complexity through algorithms, decomposition, pattern recognition, abstraction, and evaluation.		
	Systems thinking				**Understanding systems and their behaviour** Systems consist of interconnected parts that work together to transform inputs, produce outputs, and achieve a purpose.		
	Digital Technology						**Digital technologies process and communicate information** Digital systems use data and digital tools to create, process, communicate, and manage information for different purposes and users.

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Strands	Sub-elements	Year 7	Year 8	Year 9	Year 10
		Students develop fit-for-purpose technological outcomes through design, making and innovation.			
Design, Make, and Innovate	**Innovation in technology** **Designing and developing fit-for-purpose outcomes**	**Technology as responsible innovation** Technologists work with people, cultures, places, and contexts to develop responsible innovations. They use modelling, evidence, and evaluation to create fit-for-purpose outcomes while considering ethical and sustainable impacts.		**Technology as shaping futures** Technologists use specialised knowledge, evidence, and stakeholder perspectives to develop innovations that respond to complex opportunities and challenges. They balance fitness for purpose with ethical, cultural, environmental, and future-focused considerations.	
Materials Technology	**Materials Technology**	**Material choices influence function, quality, and sustainability.** Material choices and technological processes influence the function, performance, quality, and sustainability of technological outcomes.		**Materials technology involves informed and responsible material practice.** Responsible material practice requires informed decisions that balance performance, quality, longevity, sustainability, ethics, and context.	
Food and Processing Technology	**Codes of practice** **Ingredients and techniques** **Responsible design**	**Ingredients and processing shape food outcomes** Food outcomes are shaped by the interaction of ingredient properties, processing techniques, and design decisions.		**Food outcomes are developed through informed and responsible processing** Informed and responsible processing balances functionality, safety, quality, sustainability, and the needs of users and communities.	
Systems and Control	**Systems Thinking** **Physical Systems**	**Systems achieve purposes through interaction and control** System performance depends on the interaction of components, controls, and feedback that work together to achieve intended outcomes.		**Integrated systems respond, adapt and perform** Integrated systems use feedback, control, and responsive components to monitor, adapt, and optimise performance.	
Digital Technology	**Computer science** **Programming** **Digital media**	**Creating digital solutions through computational thinking** Digital solutions are created through the interaction of data, digital systems, programming, and computational thinking.		**Designing and evaluating digital innovation** Digital innovation requires the design, evaluation, and responsible use of digital technologies, data, and computational thinking.	
Spatial and Product Design	**Spatial and product design**			**Design representation shapes understanding and innovation** Design representations support the exploration, communication, and refinement of ideas by making relationships between people, products, spaces, and purposes visible.	

Conceptual Mapping: Technology (Phases 1–4)



Phase 1
Years 0–3

Design, Make, and Innovate

Phase 2
Years 4–6

Design, Make, and Innovate

Phase 3
Years 7–8

Design, Make, and Innovate

Materials Technology (Resistant and Textiles) | **Food and Processing Technology** | **Digital Technologies** | **Systems and Control**

Phase 4
Years 9–10

Design, Make, and Innovate

Spatial and Product Design | **Materials Technology (Resistant and Textiles)** | **Food and Processing Technology** | **Digital Technologies** | **Systems and Control**

Phase 5 Subjects
Year 11–13

Design, Make, and Innovate: Provides the common body of technological knowledge and practice that underpins all other strands.

Students in both Years 9 and 10 must be taught the **Design, Make, and Innovate** strand through at least two of the strands as set out in this learning area. The strands students are taught can be different in Years 9 and 10.